C++

PENG

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1 Introduction

C++ is a high-level, statically typed programming language [1].

1.1 IO

Definition 1.1 (IO objects)**.**

- `iostream` *: std lib*
- `std::cin` *:* `istream` *type*
- `std::cout` *:* `ostream` *type*

```
std::cout << "str";         // return object: std::cout
std::cout << "str" << "\n"; // (std::cout << "str") << "\n";

int v1 = 0, v2 = 0;
std::cin >> v1;            // return object: std::cin
std::cin >> v1 >> v2;      // (std::cin >> v1) >> v2;
```

Definition 1.2 (namespace)**.**

```
namespace::name // :: scope operator

// headers file should not include using declarations
using namespace::name;
```

1.2 Comments

Definition 1.3 (comment)**.**

```
// single line, best way
/* paired, can not nest */

std::cout << /* "*/" /* "/*" */;
// remove everything between */ and closest /* before
// std::cout << " /* ";
```

1.3 Flow of Control

Definition 1.4 (while)**.**

```
while(cond)
{
    statements;
}

while(std::cin >> val)
// std::cin >> val return left operand: std::cin
// istream become invalid:
// 1. reading a non-integer val: 1 2 3.1
// 2. end-of-file: Ctrl-Z then RET (windows), or Ctrl-D then RET (linux)
```

Definition 1.5 (for)**.**

```
for (init-statement; cond; expr)
{
    statements;
}

// init-statement:
// local var used only inside for, executed only once
// expr: executed after for body
```

Definition 1.6 (if)**.**

```
if (cond)
{
    statements;
}
else if (cond)
{
    statements;
}
else
{
    statements;
}

if (std::cin >> val)
    std::cout << val << "\n";
// 1 2 3 RET => 1
```

1.4 Class

Definition 1.7 (class)**.** *Define data structures:*

- *name*

- *defined*
- *operation*

```
// #include <std lib>
// #include "header.h"
```

Definition 1.8 (member function)**.**

```
data.func()
```

2 Types

Definition 2.1 (arithmetic types)**.**

```
char        // '\0': null, '\t': horizontal tab, '\n': newline
#include <string>
std::string
int         // unsigned: >= 0
bool        // true for 1 and false for 0
double
```

Definition 2.2 (object)**.** *An object is a region of memory that contain data and has a type.*

Definition 2.3 (declaration and definition)**.**

```
// define: allocate storage and defined only once
// source.cpp
int i = 10;

// declare: specify type and var names
// headers should have guards to be safe to be included multiple times
// header.h
#ifndef HEADER  // true if CLASSNAME not defined
#define HEADER  // preprocessor vars are all uppercase
extern int i;
#endif

// main.cpp
#include "header.h"
i ...
```

Definition 2.4 (scope)**.** *Delimited by* `{}`*.*

```
// vars defined outside any function body are initialized to 0
int val; // 0
int main ()
{
    int val = 1; // block scope
    // ::val == 0 global scope
    ...
}

// var: lowercase
// class: first letter uppercase
```

Definition 2.5 (bit)**.**

```
bit      combination         total         number
1 bit:   0, 1;               2     = 2^1;  0, 1
2 bits:  00, 01, 10, 11;     2*2   = 2^2;  0, 1, 2, 3
3 bits:  000, 001, ..., 111; 2*2*2 = 2^3;  0, 1, 2, 3, ..., 7
32 bits:                           = 2^32; 0, 1, 2, ..., 2^32 - 1
```

Definition 2.6 (type conversion).

```
unsigned i = -1; // 2^32 - 1 for 32-bit int

// when mix int and unsigned, int is automatically converted to unsigned
unsigned a = 1;
int b = -1;
a + b // 1 + (2^32 - 1) = 2^32 = 0
a * b // 1 * (2^32 - 1)
a < b // 1 < (2^32 - 1), always true
```

2.1 Compound Types

Definition 2.7 (reference). *A reference defines an alias for an object.*

```
int i = 0;
int &ref = i;
// ref must be bound only to an int object, not a literal
// error: int &ref = 1;
```

Definition 2.8 (pointer). *A pointer is an object that holds the address of another object.*

```
int i = 0, j = 1;
int *p, *p2;
p = &i;      // p holds the address of i
*p = 2;      // return the object i to which p points
p = nullptr; // or p = 0, null pointer, not bound to any object
p = p2;      // copy address
p2 = &j;     // p value not change

// pointer to pointer to ...
int **pp;    // * indicate pointer level
pp = &p;

// ref to pointer
int *&ref = p; // & => ref is reference, * => ref refers to pointer type
```

2.2 Qualifier

Definition 2.9 (const). *The compiler will replace the const var with its value.*

```
// const must be initialized
const int i = 1;

// const var local to a file by default
// must specify extern to be used in other files
// source.cpp
extern const i = 10;
// header.h
extern const i;
// main.cpp
#include "header.h"
i ...

// ref to const
// cannot change the object to which it bounds
const int i = 1;
const int &r = i;
const int &r2 = 2; // compiler: const int temp = 2; const int &r2 = temp;
```

```
// pointer to const
// cannot change the object to which it points
const int i = 1;
const int *p = &i;

// const pointer
// the address it holds cannot be changed
int i = 1;
int *const p = &i; // p always points to i
```

Definition 2.10 (constant expr)**.** *A constant expression is an expr whose value cannot change at compile time.*

```
constexpr int i = size();   // ok only if size is a constexpr function
constexpr int *p = nullptr; // const pointer, constexpr applies to p
```

Definition 2.11 (type alias)**.** *A synonym for a type.*

```
typedef int base;
using base = int;
typedef base *p;  // int*: pointer to int
p p1 = &i;

// const p: const modify the given type: int *const: constant pointer to int
const p p2 = 0; // int *const p2 = 0
const p *p3;    // int *const *p3
```

Definition 2.12 (const level)**.**

```
// top-level const: an object itself is const, i.e. const pointer
const int i = 1;
int *const p1 = &i;

// low-level const: point or refer to const object
const int *p2 = &i;
const int &r = i;

// when copying, top-level const is ignored,
// low-level const is never ignored
const int *const p3 = &i;
p2 = p3;
```

Definition 2.13 (auto)**.** *Deduce the type from the initializer at compile time.*

```
auto i = 1, *p = &i;   // must have initializer
auto i = 0, pi = 3.14; // error, types must be consistent

// auto may different from initializer's type
// auto ignores top-level const
```

Definition 2.14 (decltype)**.** *Return the operand type without evaluating the expr.*

```
decltype(expr) var
```

2.3 Data Structure

Definition 2.15 (structure)**.** *A data structure is a way to group together replated data elements and for using those data.*

```
// class is ordinarily defined outside functions, usually in className.h
struct className
{
    statements;
}; // must end with ;

className x, *y;
```

3 string

Definition 3.1 (string)**.** *Variable-length sequence of characters.*

```
#include <string>
// using std::string;

// class defines how objects can be initialized
std::string s = "A";   // copy ['A', '\0'], drop null char
std::string s(n, 'c'); // "c...c"

// read string
std::cin >> s1 >> s2;

// read line
std::getline(std::cin, s)

// compare
// at least one should be std::string
std::string s = "Hello";
s < "Hiya" // compare first diff char ASCII: i > e
s < "Hello World"

// concat
// cant add two string literals, i.e. "A" + "B"
s + " World\n" // "Hello World\n"

// for range statement: iterate through elems in a given sequence
for (char c : s)
    ...
for (char &c : s) // change char in string
    c = toupper(c); // uppercase

// ispunct(c): check if char is punctuation, i.e. ',', '!', ...
// isspace(c): check if char is space

// subscript operator: s[n]
// !s.empty()
// s.size(): unsigned std::string::size_type
// s.size() < -1 => 2^32 - 1 for 32-bit int
// n: unsigned std::string::size_type
// 0 <= n < s.size()
for(decltype(s.size()) i = 0; i != s.size(); i = i + 1)
    ...

// iterator
```

4 vector

Definition 4.1 (vector)**.** *Variable-length sequence of objects of a given type.*

```
#include <vector> // class template
// using std::vector;

std::vector<int> v; // empty vector
std::vector<int> v = {val_1, ..., val_n};
std::vector<int> v(size, val); // {val, ..., val}
v.push_back(elem) // append elem
```

```
// for range statement
for(auto &r : v)
    ...

// compare

// subscript operator
// !v.empty()
// v.size(): unsigned std::vector<int>::size_type
// 0, ..., v.size() - 1
```

5 Iterators

Definition 5.1 (iterator)**.** *Access the elem in container. General mechanism, all lib containers have iterator.*

```
auto b = v.begin(); // return iterator: denotes first elem
auto e = v.end();   // return iterator: denotes one after the last elem

// deref
*b // return ref to elem obj
b->mem // arrow operator: (*b).mem, i.e. b->empty()

// operation
++b // next elem
--e // previous elem
b != e

// for vector and string iterators
auto mid = b + (e - b) / 2;
b <= e
```

6 Array

Definition 6.1 (array)**.** *Array is a compound type and fixed-size container of unnamed objects of a single type that we access by position.*

```
type array_name[dim]; // dim = const expr > 0

// default-initialized array defined inside a function has undefined values

// list initialize
int a[3] = {};    // zero-initialization, all elems 0
int a2[3] = {1, 2}; // same as a[] = {1, 2, 0}, rest default to 0
std::string a3[3] = {"hi", "wolrd"}; // std::string a3[] = {"hi", "world", ""};

// char array
char a[] = "c++"; // string literal end with null char
// char a[] = {'c', '+', '+', '\0'};

// no copy assignment
// error: a = a2;

// array of pointers
int a[2] = {};
int *p[2]; // array of 2 pointers to int:
// right array size -> left elem type
```

```
// no array of ref
// error: int &r[2] = ...

// ref and pointer to array
int (*p2)[2] = &a; // p2 is a pointer to array of 2 int:
// () pointer -> right array size -> left elem type
int (&r)[2] = a;  // r refers to array of 2 int
// () ref -> right array size -> left elem type
int *(&r2)[2] = p;
// () ref -> right array size -> left elem type: pointer to int

// access elem in array
// array subscript type: size_t, unsigned
constexpr size_t array_size = 10;
int a[array_size] = {};
```

6.1 Pointers and Arrays

```
int a[] = {0, 1, 2};
int *p = a;  // int *p = &a[0];
auto p2 = a; // int *p2 = &a[0];
decltype(a) a2 = {0, 1, 2}; // int a[3] = {0, 1, 2};

// iterator
++p // &a[1]
int *e = &a[3]; // off-the-end pointer
(for int *b = a; b != e; ++b)
    ...

int *b = std::begin(a); // pointer to 1st elem
int *e = std::end(a);   // pointer to one past last elem

// arithmetic
int *e = a + 3;                        // points to one past last elem
auto n = std::end(a) - std::begin(a); // type: ptrdiff_t
int last = *(b + 2)                    // a[2]

// subscript
int *p = &a[2];
// p[1]:  *(p + 1): one past last elem
// p[-2]: *(p - 2): a[0]

// vectors and arrays
```

6.2 Arrays of Arrays

Definition 6.2 (multidimensional array)**.** *Arrays of arrays.*

```
constexpr size_t r=..., c=...;
int a[r][c] = {0};

// for range
for (auto &row : a)
    for(auto col : a)
        ...
// all control vars except innermost array must be ref

// pointers
int *(p)[c] = a;
```

```
auto *p = a;     // int (*p)[c] = &a[0];
auto q = *p;     // int *q = &a[0][0];

// type alias
using int_array = int[r];
typedef int int_array[r];
```

References

[1] S.B. Lippman, J. Lajoie, and B.E. Moo. *C++ Primer*. Addison-Wesley Professional, 2012. ISBN: 978-0-321-71411-4.